Agentic Quality Inspection Research

Executive Summary

An **Agentic Quality Inspection System** built on **MVTec AD** or **NEU-DET** is one of the strongest possible project directions for the OpenCV AI Competition 2026, which closes on October 26, 2026 at 11:59 p.m. Pacific Time (OpenCV Competition). The competition organizers explicitly list **Active Perception: Autonomous inspection with agentic orchestration or MCP** as their top suggested project area and offer a dedicated **$1,000 Agentic Vision Award** alongside the **$10,000 Overall Awards pool** ($5,000 First, $3,000 Second, $2,000 Third) and the **$1,000 Best Use of COOL Award** (OpenCV Competition).

To qualify for the Agentic Vision track and score well on the 100-point rubric, the system cannot be a static classifier paired with a chatbot that merely describes a fixed heatmap (OpenCV Rules). Instead, the visual output from **OpenCV 5** must actively change the agent's subsequent plan, tool invocation, image acquisition/processing step, or human-escalation decision (OpenCV Competition).

Competition Alignment and Rubric

Mandatory Requirements

Every eligible entry in the OpenCV AI Competition 2026 must satisfy three core technical constraints (OpenCV Competition):

1. **Substantive OpenCV 5 Runtime**: Use OpenCV 5 as a core functional component for image or video analysis, not merely for reading or resizing images before calling an external API (OpenCV Competition).
2. **Meaningful AWS Component**: Deploy a substantive part of the pipeline on Amazon Web Services, with documented reproducibility (OpenCV Competition).
3. **Working Demonstration and Evaluation**: Provide a judge-accessible web endpoint (preferred) or live demonstration, a 5-minute maximum video, an architecture diagram, and quantitative evaluation including failure cases (OpenCV Competition).

Scoring Rubrics

Rubric Track	Criterion	Weight	Project Strategy for Defect Inspection
Overall Rubric (OpenCV Rules)	Technical Execution	30%	Multi-stage OpenCV 5 pipeline (registration, CLAHE, ONNX DNN inference, contour metrology) + benchmark metrics on MVTec AD / NEU-DET (OpenCV 5 Deep Dive).

Rubric Track	Criterion	Weight	Project Strategy for Defect Inspection
Overall Rubric (OpenCV Rules)	Innovation	20%	Closed-loop agentic inspection where ambiguous anomaly scores trigger high-resolution ROI zooming, reference comparison, or illumination compensation (AgentIAD).
Overall Rubric (OpenCV Rules)	Real-World Impact	20%	Reduction in false-positive scrap rates and auditable root-cause defect reports for manufacturing QA operators (VLA-AD).
Overall Rubric (OpenCV Rules)	User Experience	10%	Interactive operator console showing visual trace steps, heatmaps, geometric measurements, and one-click override feedback (OpenCV Rules).
Overall Rubric (OpenCV Rules)	Documentation and Presentation	10%	Clear technical report, pinned dependencies, architecture diagram, and ablation study comparing single-pass vs. agentic inspection (OpenCV Competition).
Overall Rubric (OpenCV Rules)	Cloud Delivery and Operation	10%	AWS deployment (EC2 Graviton/ECS + Bedrock + S3/CloudWatch) with latency/cost telemetry and human-in-the-loop guardrails (OpenCV Rules).
Agentic Vision Award (OpenCV Rules)	Substantive OpenCV 5 + Agent	30%	Agent dynamically calls parameterized OpenCV 5 tools based on intermediate visual evidence (OpenCV Competition).
Agentic Vision Award (OpenCV Rules)	Orchestration and Autonomy	25%	Multi-step state machine / MDP loop that decides when to zoom, compare against a golden template, re-filter, or terminate (AgentIAD).
Agentic Vision Award (OpenCV Rules)	Task Effectiveness and Evaluation	20%	Quantitative proof that the multi-step agent loop reduces false positives or improves AUROC/F1 over a single-step baseline (VLA-AD).

Rubric Track	Criterion	Weight	Project Strategy for Defect Inspection
Agentic Vision Award (OpenCV Rules)	Failure Handling and Control	15%	Explicit handling of out-of-distribution/blurred images, confidence calibration, and human approval routing (OpenCV Rules).

State of the Art Literature

Recent research (2024–2026) has shifted industrial anomaly detection (IAD) from static single-pass heatmaps toward **multi-step agentic workflows** and **explainable Vision-Language Models (VLMs)**. Understanding these papers gives your project a rigorous academic foundation.

Key Architectures and Benchmarks

System / Paper	Core Mechanism	Benchmark Results	Key Takeaway for Your Build
AgentIAD (2025/2026) (AgentIAD)	Formulates defect inspection as a Markov Decision Process (MDP) using Qwen2.5-VL-3B with an adaptive tool space: **Perceptive Zoomer (PZ)** for high-res ROI cropping, **Comparative Retriever (CR)** for golden-template matching, **Web Searcher (WS)** for domain specs, and **TERMINATE** (AgentIAD).	**95.3% I-AUROC** and **97.84% binary accuracy** on MVTec AD; **98.7% I-AUROC** on VisA (AgentIAD).	Implement **Perceptive Zoomer** and **Comparative Retriever** as deterministic OpenCV 5 tools invoked only when initial anomaly confidence is ambiguous (AgentIAD).
VLA-AD (CVPRW 2026) (VLA-AD)	Three-agent pub-sub pipeline: **Perception Agent** (DINOv2 + adapter outputting score p, heatmap H, patch embeddings z), **Reasoning Agent** (Phi-4-mini generating natural-language explanations and context-aware threshold τ), and **Action Agent** (FIFO correction buffer triggering continual adaptation when false-positive/negative rate exceeds $\delta = 0.05$) (VLA-AD).	**99.7% mean I-AUROC** and **98.8% P-AUROC** across all 15 MVTec AD categories; **34.7% reduction in false positives** via adaptive thresholding (VLA-AD).	Gate LLM/VLM reasoning behind a fast perception threshold (e.g., only run deep reasoning when $p > 0.3$) and maintain an operator feedback buffer that dynamically calibrates thresholds (VLA-AD).
AnomalyGPT (AAAI 2024) (AnomalyGPT)	Feeds patch-level visual features and an anomaly localization decoder mask into an LVLM via a prompt learner, eliminating manual threshold tuning and supporting multi-turn operator dialogue (AnomalyGPT).	**94.1% I-AUROC**, **95.3% P-AUROC**, and **86.1% accuracy** on 1-shot MVTec AD (AnomalyGPT).	Pass explicit spatial localization masks and contour coordinates into the VLM prompt rather than asking a raw VLM to find tiny defects unaided (IAD-GPT).

System / Paper	Core Mechanism	Benchmark Results	Key Takeaway for Your Build
LogicAD (2025) (LogicAD)	Extracts structured text features from images using VLMs and applies a rule-based **logic reasoner** + format embedding to detect logical anomalies (missing, misplaced, or wrong-count components) (LogicAD).	**86.0% AUROC** and **83.7% F1-max** on MVTec LOCO AD, outperforming prior SOTA by 18.1% AUROC (LogicAD).	Combine pixel/texture anomaly detection with explicit geometric/count rules (e.g., component count, symmetry, pin alignment measured via OpenCV contours) (LogicAD).
MMAD Benchmark (2024/2026) (MMAD)	Full-spectrum benchmark of 39,672 questions across 8,366 images (MVTec AD, VisA, MVTec-LOCO, GoodsAD) covering 7 subtasks: anomaly discrimination, defect classification, localization, description, and root-cause analysis (MMAD).	Standalone GPT-4o reaches only **74.9% average accuracy** without visual expert tools or reference retrieval (MMAD).	Proves why standalone VLMs fail on industrial inspection and why OpenCV 5 perception + reference comparison tools are essential (Can MLLMs be Guided).

Dataset Comparison and Selection

Choosing the right dataset (or combining both in a unified interface) determines your model pipeline and the visual richness of your agentic loop.

Attribute	MVTec AD (MVTec AD Paper)	NEU Surface Defect Database (NEU-DET) (NEU-DET Benchmark)
Total Images	**5,354 high-resolution RGB images** (3,629 defect-free train/val; 1,725 mixed test) (MVTec AD Paper)	**1,800 grayscale images** (300 images per class across 6 defect classes) (NEU-DET Benchmark)
Image Resolution	700×700 **to** 1024×1024 **pixels** (acquired with 2048×2048 RGB sensors and telecentric lenses) (MVTec AD Docs)	200×200 **pixels**, 8-bit grayscale (Cross-Dataset Benchmarking)
Categories & Defect Types	**15 categories** (10 objects: bottle, cable, capsule, hazelnut, metal nut, pill, screw, toothbrush, transistor, zipper; 5 textures: carpet, grid, leather, tile, wood) with **73 fine-grained physical defect types** (MVTec AD Docs)	**1 material (hot-rolled steel strip)** with **6 defect classes**: crazing (Cr), inclusion (In), patches (Pa), pitted surface (PS), rolled-in scale (RS), and scratches (Sc) (NEU-DET Benchmark)
Annotations Provided	**1,888 pixel-precise binary ground-truth masks** for anomalous test regions (MVTec AD Paper)	**Pascal VOC XML bounding boxes** (NEU-DET) and pixel masks in the NEU-Seg subset (Datadory NEU-DET)
Primary ML Paradigm	**Unsupervised / One-Class Anomaly Detection** (trained only on defect-free images; detects unseen structural and logical anomalies) (MVTec Methods)	**Supervised Object Detection / Classification** (trained on labeled defect boxes via YOLOv8/YOLOX/RT-DETR) (Steel Surface Defect Detection)

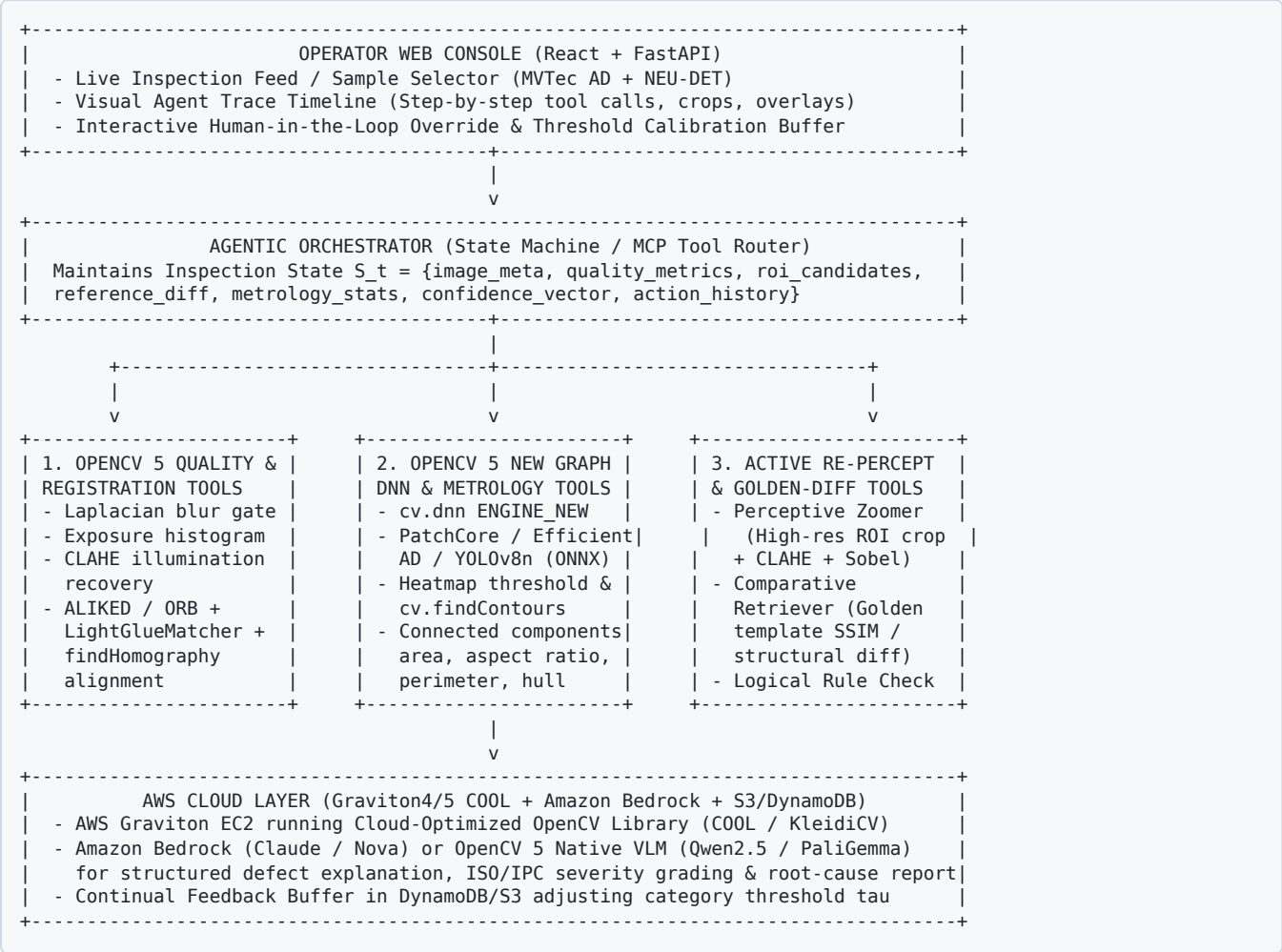
Attribute	MVTec AD (MVTec AD Paper)	NEU Surface Defect Database (NEU-DET) (NEU-DET Benchmark)
Suitability for Agentic Loop	**Exceptional**: High resolution (1024×1024) makes **Perceptive Zooming** meaningful; defect-free training templates enable **Golden-Template Alignment & Differencing**; object categories (transistor, cable, metal nut, bottle) allow **Logical/Geometric Rule Verification** (AgentIAD).	**Moderate as standalone, strong as secondary**: Fast to train a lightweight YOLOv8n ONNX detector via OpenCV 5 DNN, and great for **Morphological Metrology** (crack length, pit density, scratch orientation), but 200×200 grayscale limits multi-scale zoom inspection (Paper1 Edge Defect Detection).

Recommendation

Use **MVTec AD** (focusing on 4–5 high-impact categories such as `bottle`, `transistor`, `metal_nut`, `cable`, and `tile` or `wood`) as your **primary dataset** to showcase unsupervised anomaly localization, golden-template registration, logical component verification, and high-resolution ROI zooming (MVTec AD Paper). Optionally include **NEU-DET** as a second **"High-Speed Steel Strip Mode"** powered by a YOLOv8n ONNX model in OpenCV 5 DNN (`10.9 ms` CPU inference in OpenCV 5) to demonstrate that your agentic orchestrator handles both **unsupervised anomaly maps** and **supervised multi-class bounding boxes** (OpenCV 5 Deep Dive).

System Architecture for OpenCV 5

To win both high Overall Rubric marks and the Agentic Vision Award, every stage of the perception–decision–action loop should invoke concrete **OpenCV 5 capabilities** (OpenCV 5 Deep Dive).



Multi-Step Perception–Decision–Action Loop

Here is how visual evidence from OpenCV 5 dynamically alters the agent's execution path at every step, directly satisfying the Agentic Vision Award requirement (OpenCV Competition):

Step 1: Pre-Inspection Gatekeeper (OpenCV 5 Image Quality & Photometric Check)

- **OpenCV 5 Tool**: Computes variance of the Laplacian (`cv.Laplacian`) for focus/blur, intensity histogram quantiles (`cv.calcHist`) for specular glare or underexposure, and signal-to-noise ratio.
- **Agent Decision**:
  - If severely blurred ($\mathrm{Var}(\nabla^2 I) < \tau_{\text{blur}}$): **Halt DNN inference** and emit `ACTION: REQUEST_RETAKE` with a localized blur map so the system never hallucinates defects on out-of-focus frames.
  - If uneven lighting or low contrast is detected (common in steel surfaces or MVTec AD2 dark-field scenarios): **Dynamically invoke** `Tool: Photometric_Normalize` using `cv.createCLAHE` in LAB color space and bilateral filtering (`cv.bilateralFilter`) before passing to the detector (MVTec AD 2).

Step 2: Coarse Anomaly Localization (OpenCV 5 Graph DNN Engine)

- **OpenCV 5 Tool**: Runs an ONNX anomaly/defect model via `cv.dnn.readNetFromONNX` using OpenCV 5's rebuilt graph engine (`ENGINE_NEW`), which supports over 80% of ONNX operators, dynamic shapes, and fused attention (`24.4%` faster than ONNX Runtime on DINOv2-small and `11.5%` faster on YOLOv8n) (OpenCV 5 Deep Dive).
  - For **MVTec AD**: Uses an ONNX-exported `PatchCore` or `EfficientAD` model (via Anomalib) producing a global anomaly score $p \in [0, 1]$ and pixel anomaly map $H \in \mathbb{R}^{H \times W}$ (Manufacturing

Anomaly Guide).
    - For **NEU-DET**: Uses an ONNX-exported `YOLOv8n` model producing candidate bounding boxes and class probabilities across the 6 steel defect types (OpenCV 5 Deep Dive).
- **Agent Decision**:
    - **Fast-Path Pass** ($p < \tau_{\text{low}}$, e.g., $p < 0.25$): Immediately log as `PASS` without wasting VLM tokens, achieving sub-50 ms throughput (VLA-AD).
    - **Ambiguous / Micro-Defect Path** ($\tau_{\text{low}} \le p < \tau_{\text{high}}$, e.g., $0.25 \le p < 0.65$, or candidate contour area $< 0.5\%$ of image): The agent **cannot decide yet** and branches into **Step 3 (Active Re-Perception)** (AgentIAD).
    - **High-Confidence Defect Path** ($p \ge \tau_{\text{high}}$): Proceeds directly to **Step 4 (Metrology & Explainable Grading)**.

Step 3: Active Re-Perception for Ambiguous Cases (Zoom + Golden-Template Alignment)

When initial evidence is uncertain, the agent selects from three specialized OpenCV 5 tools based on product category and spatial features: 1. `Tool: Perceptive_Zoomer` **(High-Resolution ROI Re-Analysis)**: Crops the top-k candidate bounding boxes from the full-resolution 1024×1024 image with padding, applies unsharp masking + `cv.Canny` / `cv.Sobel` gradient magnitude analysis, and re-evaluates local texture anomalies (AgentIAD). 2. `Tool: Comparative_Golden_Retriever` **(Feature Registration & Structural Differencing)**: Retrieves the nearest defect-free "golden template" from the MVTec training split, aligns the query image to the template using OpenCV 5's new learned feature matchers (`ALIKED` / `DISK` + `LightGlueMatcher`, or `ORB`) and `cv.findHomography` with RANSAC, warps the template via `cv.warpPerspective`, and computes a masked structural residual map (OpenCV 5 Deep Dive). This separates true structural anomalies (missing transistor leg, chipped bottle neck, bent cable wire) from benign surface dust or pose shifts. 3. `Tool: Logical_Geometry_Checker`: Uses `cv.connectedComponentsWithStats`, `cv.moments`, and `cv.matchShapes` to verify topological constraints (e.g., whether a transistor has all expected leads in the right bounding zones, or whether a metal nut has valid inner-thread circularity via `cv.HoughCircles`) (LogicAD).

Step 4: Quantitative Metrology and Explainable VLM Synthesis

- **OpenCV 5 Metrology Tool**: Converts the verified anomaly mask into objective engineering measurements using `cv.findContours`, `cv.contourArea`, `cv.arcLength`, `cv.minAreaRect`, and `cv.fitEllipse`:
    - Defect bounding box (x, y, w, h), pixel area, percentage of component surface area, elongation/aspect ratio (distinguishing linear scratches/cracks from circular pits/inclusions), and minimum distance to critical functional edges.
- **Structured VLM Reasoning (AWS Bedrock or OpenCV 5 Native VLM)**:
    - Passes the **annotated multi-panel evidence** (Original, Registered Golden Diff, Zoomed ROI, and JSON Metrology Table) to a Vision-Language Model (Amazon Bedrock Claude/Nova, or a local Qwen2.5/PaliGemma model via OpenCV 5's native DNN tokenizer and KV-cache) (OpenCV 5 Deep Dive).
    - Because the VLM is grounded in exact OpenCV measurements rather than raw pixels alone, it generates an **auditable quality inspection report**: defect classification, physical morphology explanation, severity grade (`PASS`, `REWORK`, `SCRAP`, or `HUMAN_REVIEW`), and recommended corrective action (VLA-AD).

Step 5: Continual Threshold Calibration & Human-in-the-Loop Action

- Inspired by **VLA-AD**, when an operator marks a borderline flag as a false positive or confirms a defect in the dashboard, the **Action Agent** logs the feature embedding and outcome to a feedback buffer (VLA-AD).
- If the false-positive rate on a category rises above a threshold (e.g., due to a lighting shift), the agent automatically recalibrates the decision threshold τ_{category} or triggers the illumination-normalization pre-filter by default on subsequent frames (VLA-AD).

Targeting Both Special Awards

You can make your project eligible for **both** $1,000 Special Awards in addition to the $10,000 Overall Prize pool (OpenCV Competition):

1. **Agentic Vision Award ($1,000)**:
   - Include an explicit **Agent Workflow Diagram** in your technical report and UI (OpenCV Competition).
   - Log and display the **JSON Execution Trace** for every inspected part showing how OpenCV 5 outputs (e.g., Laplacian blur score, ambiguous PatchCore score $p = 0.42$, homography alignment residual) directly triggered the next tool call (`Photometric_Normalize`, `Perceptive_Zoomer`, `Comparative_Golden_Retriever`, or `Request_Human_Approval`) (OpenCV Competition).
2. **Best Use of COOL Award ($1,000)** (Optional High-ROI Add-On):
   - Deploy your Python/FastAPI OpenCV 5 backend on an **AWS Graviton4 (`c8g` / `m8g`) or Graviton5 (`c9g`) EC2 instance** using the **Cloud-Optimized OpenCV Library (COOL)** from AWS Marketplace, which accelerates resize, Gaussian blur, adaptive thresholding, and contour detection by **20% to 70%** via Arm KleidiCV vector instructions (AWS COOL Blog).
   - Run a simple benchmark script on 200 MVTec AD / NEU-DET images comparing **Standard OpenCV vs. COOL on AWS Graviton** (measuring preprocessing + registration + contour extraction latency in milliseconds) and include that table in your report (OpenCV Rules).

Evaluation Plan for Judges

The competition rubric awards 30% for Technical Execution and 20% in the Agentic Vision rubric for Task Effectiveness and Evaluation, explicitly requiring failure-case analysis (OpenCV Rules). Structure your evaluation section around three concrete tables:

1. **Ablation Study (Single-Pass Baseline vs. Agentic Loop)**:
   - Compare **Baseline** (single-pass ONNX detector at fixed threshold $\tau = 0.5$) against **Agentic Pipeline** (Quality Gate + Golden-Template Diff + Perceptive Zoomer + Context-Calibrated Threshold) on a held-out test split of MVTec AD (e.g., 300 images across `bottle`, `transistor`, `metal_nut`, `cable`, `wood`) and/or NEU-DET (300 test images) (VLA-AD).
   - Report **Image-AUROC**, **F1 Score**, **False Positive Rate (FPR) on normal parts**, and **End-to-End Latency (ms)** (showing how the fast-path skips expensive VLM calls on clean parts) (VLA-AD).
2. **Robustness Under Simulated Factory Corruptions**:
   - Following VLA-AD's drift benchmark, evaluate how the agent's OpenCV quality gate and photometric recovery perform when test images are degraded with synthetic motion blur, gamma shifts ($\gamma = 1.5$), or Gaussian noise (VLA-AD).

3. **Documented Failure Cases**:
    - Explicitly show 3–4 failure modes required by the rules: (a) non-planar 3D rotation where 2D homography (`cv.findHomography`) leaves edge alignment artifacts, (b) ultra-low-contrast micro-cracks on NEU-DET `crazing` samples that blend with rolling grain texture, and (c) logical anomalies where texture appears normal (OpenCV Competition).

Ten-Day Execution Roadmap

With the final deadline on **October 26, 2026** (OpenCV Competition), here is a practical build schedule:

- **Days 1–2 (Core OpenCV 5 Tooling & Models)**: Set up OpenCV 5 environment; export pre-trained `PatchCore`/`EfficientAD` ONNX weights (via Anomalib) for 4–5 MVTec AD categories and/or `YOLOv8n` ONNX for NEU-DET (Manufacturing Anomaly Guide). Implement the deterministic OpenCV 5 tools (`check_image_quality`, `normalize_illumination`, `run_dnn_detector`, `align_and_diff_golden_template`, `zoom_and_measure_contours`).
- **Days 3–4 (Agentic State Machine & VLM Explainability)**: Build the multi-step orchestrator in Python (FastAPI) where intermediate OpenCV scores trigger tool calls (`Perceptive_Zoomer`, `Comparative_Golden_Retriever`, `Request_Retake`, or `Escalate_Human_Review`) and connect Amazon Bedrock (or OpenCV 5 native VLM) to synthesize structured JSON inspection reports (AgentIAD).
- **Days 5–6 (Operator Web Console)**: Build a focused 3-panel React frontend: (1) Part/Sample Selector & Image Upload with simulated factory corruptions (blur/glare toggle), (2) Visual Agent Trace showing each OpenCV 5 tool output side-by-side (Original, Homography-Aligned Golden Diff, High-Res Zoomed Contour Overlay), and (3) Explainable Engineering Report + Operator Feedback button (OpenCV Competition).
- **Days 7–8 (AWS Deployment & Quantitative Evaluation)**: Deploy on AWS (ideally an AWS Graviton instance with COOL for the bonus track, plus Bedrock/S3) and run the automated evaluation script comparing Single-Pass vs. Agentic Pipeline across held-out MVTec AD / NEU-DET test sets (AWS COOL Blog).
- **Days 9–10 (Technical Report, Diagrams & 5-Minute Video)**: Write the competition technical report covering all 7 required sections, generate the architecture and agent-workflow diagrams, and record the 5-minute demonstration video (OpenCV Competition).